

1. What is a logical agent?

- A logical agent uses a knowledge base of logical sentences to decide what to do.
- It reasons about what is true in the world and chooses actions that follow from the KB.
- Actions are selected by logical inference, not just from the current percept.

Key Idea: Represent what the agent knows in logic and use inference to determine what actions are entailed.

4. Representation and reasoning

- Knowledge is represented in a formal logical language (e.g. propositional logic, first-order logic).
- Reasoning is done using inference rules to derive new sentences from existing ones.
- The agent can deduce facts that are not directly observed.
- The representation should capture the relevant aspects of the environment in a simple and useful way.

Example: Propositional inference

KB: Raining $\rightarrow$ Wet
Raining
—
 $\vdash$ Wet
(Modus Ponens)

From KB we can infer that the ground is wet.

7. Benefits of logic in AI

- Provides a clear and precise knowledge representation.
- Supports reasoning to infer new facts.
- Allows handling of partial and indirect information.
- Can represent general knowledge about many situations.
- Separates knowledge from the control strategy.
- Flexible – new knowledge can be added easily.
- Foundation for more advanced forms of reasoning (e.g. planning, uncertainty).

9. Key definitions

Logical agent	An agent that uses logical inference over a knowledge base to select actions.
Knowledge base (KB)	A set of logical sentences representing what the agent knows.
Sentence	A well-formed formula in a logical language.
Query	A question about what follows from the KB.
TELL(KB, α)	Add the sentence α to the KB.
ASK(KB, α)	Return the logical consequences of KB with respect to α .
Inference	A process of deriving new sentences from existing ones.
Knowledge-based agent	An agent that maintains a KB and uses inference to act.

2. Knowledge-based agents

- A knowledge-based agent (KBA) maintains an explicit knowledge base (KB) of facts about the world.
- It uses inference to derive new information and to choose actions.
- More flexible and general than reflex agents because it can reason about unobserved facts.

Example: A robot knows facts about its location, obstacles and goals, and infers a safe action to take.

5. Agent program idea

- The agent program maintains a KB and uses it to decide an action at each time step.
- On each new percept, it adds (tells) relevant information to the KB, queries the KB to find out what action should be taken, and then executes the action.

Simple outline:

- Initialize KB with general knowledge.
- Upon each new percept p :
 - TELL(KB, new information from p).
 - action $\leftarrow$ ASK(KB, what to do).
 - Execute action.
 - TELL(KB, the action taken and its effects).
- Repeat for the next percept.

8. Simple real-world examples

Problem	KB (example facts)	Query / Goal	Action
Wumpus World	At(1,1), $\rightarrow$ Breeze(1,1), $\rightarrow$ Stench(1,1)	Is Safe(1,2)?	Move to (1,2)
Route finding	Road(A,B), Road(B,C), $\rightarrow$ Blocked(B,C)	Is there a route to C?	Move to B (or C)
Robot at home	At(Kitchen), Battery(High)	Find location of Mug	Move to LivingRoom
Game playing	PlayerTurn, Legal(Move1), Legal(Move2)	What move is best?	Make move
Scheduling	Free(9am), Meeting(10am)	What can I do now?	Do Task

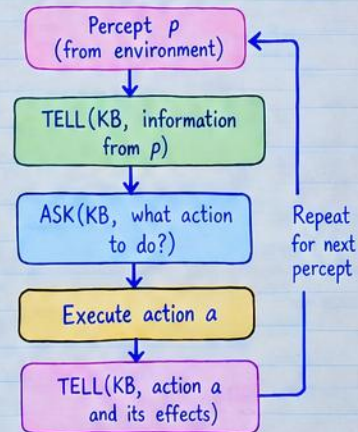
3. Knowledge base (KB), sentences, queries, tell and ask

Knowledge Base (KB)

- A set of logical sentences (formulas) representing what the agent knows.
- Sentences describe facts about the world (e.g. At(Loc1), Clear(Loc2), $\rightarrow$ Breeze(1,1)).

Sentence	A well-formed formula in a formal language (e.g. propositional or first-order logic).
Query	A question about what follows from the KB (e.g. "Is Safe(1,2)?").
TELL(KB, α)	Add a new sentence α to the KB.
ASK(KB, α)	Return the logical consequences of KB with respect to query α (e.g. true/false or all entailed answers).

6. Small knowledge-based agent cycle



10. Summary

- Logical agents maintain a knowledge base of logical sentences.
- They use inference to decide what actions to take.
- Knowledge is represented in a formal language.
- The main operations are TELL and ASK.
- The agent program adds percepts to the KB, queries for an action, executes it, and repeats.
- Logical agents are more flexible and general than simple reflex agents.
- Logic provides a clear and powerful framework for knowledge representation and reasoning in AI.

“A logical agent reasons with what it knows to decide what to do.”

1. Wumpus World Overview

- A simple, fictional environment used to demonstrate how logical agents can reason about their environment.
- The agent explores a grid of caves, trying to find a gold piece while avoiding pits and a monster called the Wumpus.
- The agent has only partial and local information (percepts) and must use logical inference to be safe.
- Illustrates knowledge representation, reasoning and action selection.

Key Idea: Use logic to infer what is in unobserved squares based on percepts from visited squares.

4. Why Wumpus World is Useful for Logical Reasoning

- Simple but non-trivial environment.
- Agent has only partial information.
- Requires inference to deduce hidden facts about pits and the Wumpus.
- Demonstrates knowledge-based agents and logical reasoning.
- Illustrates how to derive new knowledge from percepts using logical rules.

A classic example to show how logic helps an agent act rationally under uncertainty (partial observability).

2. Environment Properties

Type	Single agent
Observability	Partially observable (only local percepts)
Deterministic	Yes (outcomes are certain)
Episodic or Sequential	Sequential
Static or Dynamic	Static (doesn't change while thinking)
Discrete or Continuous	Discrete (finite grid)
Known or Unknown	Known rules, but initial state unknown
Environment size	Cave of $N \times N$ squares (e.g., 4×4 typical)

3. Wumpus World Symbols

Pit (P)		A bottomless pit. If agent enters, it falls and dies.
Breeze (B)		Percept felt in squares adjacent to a pit.
Stench (S)		Percept felt in squares adjacent to the Wumpus.
Glitter (G)		Percept in the square containing gold.
Gold		A single piece of gold. Goal is to grab it.
Wumpus (W)		A monster that eats the agent if it enters the same square.
Agent (A)		The explorer that moves around, senses and acts.
Safe square		A square known (or inferred) not to contain a pit or the Wumpus.

5. Example 4×4 Wumpus World Grid

	1	2	3	4
4				
3				
2				
1				

This is just one possible configuration. The agent does not know the true locations initially.

Legend:

	= Agent
	= Wumpus
	= Pit
	= Gold
	= Breeze (adjacent to pit)
	= Stench (adjacent to Wumpus)
	= Safe square (known safe)

6. Percepts and Actions

Percepts (from current square)	
Stench (S)	Wumpus in an adjacent square.
Breeze (B)	Pit in an adjacent square.
Glitter (G)	Gold in the current square.
No percept	No stench, no breeze, no glitter (i.e., likely safe, but not certain).

Actions (what the agent can do)	
Move Forward	Move to the square in front.
Turn Left	Rotate 90° left.
Turn Right	Rotate 90° right.
Grab	Pick up the gold from current square.
Shoot	Fire an arrow in the current direction (to kill the Wumpus).
Climb	Exit the cave (usually from start square).

8. Safe Inference Idea

- Start from a square with no breeze and no stench.
- Infer that all adjacent squares are safe (no pit and no Wumpus).
- Use logical rules repeatedly to infer more safe squares.
- Build a knowledge base with all known facts.
- Only move to squares that are inferred to be safe.

Example: If the start square (1,1) has no breeze and no stench, then (1,2) and (2,1) are safe. From newly visited safe squares, infer further safe squares using percepts.

9. Agent Goals

- Find the gold.
- Return to the start square and climb out.
- Avoid falling into a pit.
- Avoid being eaten by the Wumpus.
- Use logical inference to choose safe moves and potentially shoot the Wumpus.

Success condition:

Grab the gold and climb out alive (without falling into a pit and without being eaten by the Wumpus).

7. Sample Logical Rules

- R1. Breeze rule:**
 $B(x,y) \rightarrow (P(x+1,y) \vee P(x-1,y) \vee P(x,y+1) \vee P(x,y-1))$
 If there is a breeze in (x,y) , then at least one adjacent square has a pit.
- R2. Stench rule:**
 $S(x,y) \rightarrow (W(x+1,y) \vee W(x-1,y) \vee W(x,y+1) \vee W(x,y-1))$
 If there is a stench in (x,y) , then at least one adjacent square has the Wumpus.
- R3. No breeze:**
 $\neg B(x,y) \rightarrow \neg P(x',y')$ for all adjacent (x',y')
 If there is no breeze, then none of the adjacent squares have a pit.

10. Key Takeaways

- Wumpus World is a classic testbed for logical agents.
- The environment is partially observable.
- Percepts provide local information (stench, breeze, glitter).
- Logical rules let the agent infer hidden facts (pits, Wumpus, safe squares).
- Safe inference allows the agent to explore without taking unnecessary risk.
- The agent's goal is to find the gold and exit safely.
- Wumpus World illustrates how logic enables rational action under uncertainty.

1. Syntax and Semantics in Logical Representation

- Syntax:** rules for forming well-formed sentences (WFFs) in a formal language.
- Semantics:** meaning of the sentences, defined in terms of models (interpretations) and truth.
- We use propositional logic first: sentences are built from propositional symbols and logical connectives.

Key Idea: A logical sentence has a syntactic form (how it is written) and a semantic meaning (its truth in a model).

3. Logical Connectives

Connective	Symbol	Meaning	Formula (example)
NOT	$\neg$	Negation (not)	$\neg P$
AND	$\wedge$	Conjunction (and)	$P \wedge Q$
OR	$\vee$	Disjunction (or)	$P \vee Q$
IMPLIES	$\Rightarrow$	Implication (if ... then)	$P \Rightarrow Q$
IFF	$\Leftrightarrow$	Biconditional (if and only if)	$P \Leftrightarrow Q$

5. Models and Truth

- A **model** (interpretation) assigns a truth value (True or False) to each propositional symbol.
- The truth value of a complex sentence is determined by the truth values of its parts and the meaning of the connectives.
- A sentence is true in a model if it evaluates to True under that interpretation.

Example: In a model where $P = \text{True}$ and $Q = \text{False}$:

- $P \wedge Q$ is False
- $P \vee Q$ is True
- $\neg Q$ is True
- $P \Rightarrow Q$ is False (since $P = \text{True}$ and $Q = \text{False}$)

7. Validity, Satisfiability, Contradiction

- Valid (tautology):** A sentence is valid if it is true in all models.
Example: $P \vee \neg P$ is valid.
- Satisfiable:** A sentence is satisfiable if it is true in at least one model.
Example: $P \wedge Q$ is satisfiable.
- Contradiction:** A sentence is a contradiction if it is false in all models.
Example: $P \wedge \neg P$ is a contradiction.

9. Examples of Logical Sentences from Wumpus World

Statement (English)	Logical Sentence
There is a pit in (1,1).	$P_{1,1}$
There is a wumpus in (1,1).	$W_{1,1}$
There is a breeze in (1,1) iff a pit is adjacent.	$B_{1,1} \Leftrightarrow (P_{1,2} \vee P_{2,1})$
There is a stench in (1,1) iff a wumpus is adjacent.	$S_{1,1} \Leftrightarrow (W_{1,2} \vee W_{2,1})$
It is safe to move to (1,1).	$OK_{1,1} \Leftrightarrow \neg P_{1,1} \wedge \neg W_{1,1}$
Agent is at (1,1) and perceives a breeze.	$L_{1,1} \wedge B_{1,1}$

2. Propositional Symbols

- Propositional symbols (also called atomic propositions) represent simple facts about the world.
- Examples: $P, Q, R, W_{11}, \text{Breezy}, P_{1,2}$, etc.
- Each symbol is either True or False in a given model (interpretation).
- Symbols can stand for complex facts, e.g. $P_{1,2}$: "There is a pit in cell (1,2)".

Example: In the Wumpus World:

$P_{1,1}$: There is a pit in (1,1).	$B_{1,1}$: There is a breeze in (1,1).
$W_{1,1}$: There is a wumpus in (1,1).	$G_{1,1}$: There is gold in (1,1).
$S_{1,1}$: There is a stench in (1,1).	$OK_{1,1}$: It is safe to move to (1,1).

4. Well-Formed Sentences (WFFs)

- A propositional symbol is a WFF.
- If A is a WFF, then $\neg A$ is a WFF.
- If A and B are WFFs, then $(A \wedge B)$, $(A \vee B)$, $(A \Rightarrow B)$, $(A \Leftrightarrow B)$ are WFFs.
- Use parentheses to make the structure clear.

Examples of WFFs:

- P
- $\neg P$
- $(P \wedge Q)$
- $(P \vee (Q \wedge R))$
- $(P \Rightarrow Q)$
- $((P \vee Q) \Leftrightarrow \neg R)$

Not WFFs (ill-formed):

- $P \wedge$ (missing rhs)
- $\neg(P \wedge Q$ (missing parenthesis)
- $P Q$ (missing connective)

6. Entailment

- A knowledge base KB entails a sentence α (written $KB \models \alpha$) if α is true in every model where KB is true.
- Entailment captures the idea of logical consequence.
- $KB \models \alpha$ means α follows from KB; α is true in all models that satisfy KB.

Example:

$KB = \{P \Rightarrow Q, P\}$ entails Q ($KB \models Q$)
Because in every model where $P \Rightarrow Q$ and P are true, Q must also be true.

8. Truth Table Illustration

Truth table for $(P \Rightarrow Q)$ and $(P \Leftrightarrow Q)$:

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
T	F	F	T	T	F	F
T	T	F	T	T	T	T
F	T	T	F	T	T	F
F	F	T	F	F	T	T

Note: $P \Rightarrow Q$ is false only when P is true and Q is false.
 $P \Leftrightarrow Q$ is true when P and Q have the same truth value.

10. Summary

- Logical sentences have syntax (form) and semantics (meaning).
- Propositional symbols represent basic facts (True/False).
- Use logical connectives: $\neg, \wedge, \vee, \Rightarrow, \Leftrightarrow$.
- Build well-formed sentences (WFFs) using syntactic rules.
- A model assigns truth values to symbols; semantics defines truth.
- Entailment ($KB \models \alpha$) means α is true in all models of KB.
- Valid: true in all models. Satisfiable: true in some model. Contradiction: false in all models.
- Truth tables show the meaning of connectives.
- Logical sentences can represent properties of the Wumpus World.
- These foundations let logical agents reason about the world.

1. Inference in Knowledge Bases

- A knowledge base (KB) contains sentences about the world (facts, rules, percepts).
- Inference is the process of deriving new sentences (facts) from the KB.
- If $KB \models \alpha$ (KB logically entails α), then α is true in all models where KB is true.
- Logical inference lets an agent deduce new information, not just store facts.

Key Idea:

Use inference rules (based on logic) to derive new knowledge from existing knowledge in the KB.

2. Soundness and Completeness

Soundness

- An inference procedure is sound if every sentence it derives is logically entailed by the KB (i.e., it never derives false conclusions).
- Sound $\Rightarrow$ No incorrect conclusions.

Completeness

- An inference procedure is complete if it can derive every sentence that is logically entailed by the KB.
- Complete $\Rightarrow$ Can derive all true conclusions (though it may be slow).

Ideally, we want inference procedures that are both sound and complete.

3. Model Checking Idea

- Model checking tests whether $KB \models \alpha$ by enumerating all possible models (world states).
- Check if in every model where KB is true, α is also true.
- If there is a model where KB is true and α is false, then $KB \not\models \alpha$.

Algorithm (Model Checking)

- For each possible model m :
 - If m satisfies KB and not α , then return FALSE ($KB \not\models \alpha$).
- If no such model is found, return TRUE ($KB \models \alpha$).

Model checking is simple but exponential in the number of symbols, so it is not practical for large KBs.

4. Logical Inference Examples in Wumpus World

- Use propositional logic to represent what the agent perceives and what it knows.
- Rules relate percepts to nearby hazards (breeze $\leftrightarrow$ pit, stench $\leftrightarrow$ wumpus).

3	(1,3)	(2,3)	(3,3)
2	(1,2)	(2,2)	(3,2)
1	S (1,1)	(2,1)	(3,1)
	1	2	3

Knowledge base for Wumpus World (simplified):

- $B(x,y) \leftrightarrow (P(x+1,y) \vee P(x-1,y) \vee P(x,y+1) \vee P(x,y-1))$
- $S(x,y) \leftrightarrow (W(x+1,y) \vee W(x-1,y) \vee W(x,y+1) \vee W(x,y-1))$

Initial percept at (1,1): no breeze, no stench.

- $\neg B(1,1) \Rightarrow \neg P(2,1) \wedge \neg P(1,2)$
(no pits adjacent)
- $\neg S(1,1) \Rightarrow \neg W(2,1) \wedge \neg W(1,2)$
(no wumpus adjacent)

5. Forward Chaining vs Backward Chaining

Forward Chaining

- Start with known facts (KB).
- Repeatedly apply inference rules to derive new facts.
- Data-driven (bottom-up).
- Useful when we want to derive many possible conclusions (e.g., explore what is known).

Idea: Keep adding new facts until no new facts can be derived.

Backward Chaining

- Start with a query (goal).
- Work backwards, looking for rules that could derive the query.
- Goal-driven (top-down).
- Useful when we want to prove a specific goal (e.g., "Is (2,2) safe?").

Idea: Break the goal into subgoals and try to prove them using known facts.

6. Resolution Idea (Basic Level)

- Resolution is a single inference rule for propositional logic that is sound and complete.
- Works with sentences in Conjunctive Normal Form (CNF) (AND of ORs of literals).
- If we can derive a contradiction (empty clause $\square$), then $KB \models \alpha$.

Resolution Rule:

$$\frac{(A \vee B) \quad (\neg A \vee C)}{B \vee C}$$

Example:

$$\frac{(P \vee Q) \quad (\neg P \vee R)}{Q \vee R}$$

- We repeatedly apply resolution to pairs of clauses to derive new clauses.
- If we eventually derive the empty clause ($\square$), we have a contradiction.

7. Example: Deriving a Safe Square in Wumpus World

Suppose the agent is at (1,1) and perceives no breeze and no stench.

Knowledge base (relevant parts):

- $\neg B(1,1)$
- $\neg S(1,1)$
- $B(1,1) \leftrightarrow (P(2,1) \vee P(1,2))$
- $S(1,1) \leftrightarrow (W(2,1) \vee W(1,2))$

Inference steps:

- From 1 and 3: $\neg B(1,1) \Rightarrow \neg P(2,1) \wedge \neg P(1,2)$
- From 2 and 4: $\neg S(1,1) \Rightarrow \neg W(2,1) \wedge \neg W(1,2)$
- Therefore, (2,1) and (1,2) have no pit and no wumpus.

Conclusion:

- Squares (2,1) and (1,2) are safe to visit.
- The agent can move to either (2,1) or (1,2).
- Use this new knowledge to continue exploring the world (e.g., perceive from the new square).

3	?	?	?
2	Safe (1,2)	?	?
1	S (1,1)	Safe (2,1)	?
	1	2	3

8. Simple Algorithm / Reasoning Steps

Logical Agent with Inference (sketch)

- Initialize KB with initial knowledge (e.g., rules about the world).
- At each time step:
 - Add the new percepts to the KB.
 - Use an inference procedure (e.g., forward or backward chaining, or resolution) to derive new facts.
 - Update the set of safe, possible, and unsafe squares.
 - Choose an action based on the inferred knowledge (e.g., move to a safe and unvisited square).
- Repeat until the goal is achieved (e.g., find the gold or finish).

9. Why Inference Matters

- Lets an agent deduce new facts from limited percepts.
- Helps plan safe actions (e.g., avoid pits and wumpus in Wumpus World).
- Allows reasoning about unknown parts of the environment.
- Essential for building intelligent agents that can act in complex, uncertain environments.

Inference turns raw percepts into useful knowledge for decision making.

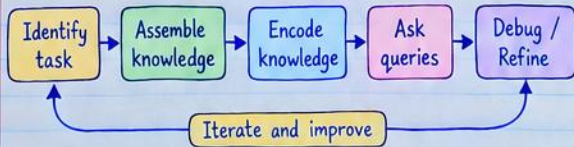
10. Key Takeaways

- Inference derives new sentences from a knowledge base.
- Soundness = no false conclusions.
- Completeness = can derive all true conclusions.
- Model checking is simple but often not practical.
- Forward chaining is data-driven; backward chaining is goal-driven.
- Resolution is a sound and complete inference rule (propositional logic).
- Inference helps a Wumpus World agent identify safe squares.
- Logical inference is crucial for intelligent behaviour in complex worlds.

1. Knowledge Engineering Process

- Knowledge engineering is the process of building a knowledge base for a logical agent so that it can reason and solve problems.
- Involves identifying what the agent needs to know, encoding it in a formal language, and refining it through testing.

Key Idea: Good knowledge engineering is crucial for building effective logical agents.



3. Applications of Logical Agents

- Robotics: planning, navigation, manipulation.
- Natural language understanding and question answering.
- Expert systems (medical diagnosis, troubleshooting).
- Planning and scheduling (logistics, timetables).
- Game playing (e.g., using logical rules and inference).
- Semantic web and knowledge graphs.
- Legal and policy reasoning.
- Configuration and design (e.g., hardware/software systems).
- Intelligent tutoring systems.

5. Reactive Agents vs. Knowledge-Based Agents

Aspect	Reactive Agents	Knowledge-Based Agents
Basis	Condition-action rules	Explicit knowledge + inference
Internal state	Usually none	Maintains knowledge base
Reasoning	No explicit reasoning	Uses logical inference
Handling new situations	Limited (predefined rules)	Can infer from knowledge
Flexibility	Less flexible	More flexible and general
Example	Vacuum cleaner agent	Logical agent for planning, diagnosis, question answering

7. Mini Glossary

Term	Meaning
Knowledge engineering	Process of building a knowledge base.
Domain knowledge	Facts and rules about a specific domain.
Knowledge base (KB)	Set of sentences (facts and rules).
Axiom	A general rule in the KB.
Fact	A specific piece of knowledge.
Inference	Deriving new sentences from the KB.
Query	A question asked to the KB.
Entailment ($\models$)	KB entails α if α is true in all models of KB.
Model	An interpretation where sentences are true.
Debugging	Finding and fixing errors in the KB.

2. Steps in the Knowledge Engineering Process

- Identify the task**
 - Define the problem and performance measure.
 - Decide what the agent needs to achieve.
- Assemble relevant knowledge**
 - Gather domain knowledge from experts, textbooks, databases, or observation.
 - Identify useful concepts, objects, relations and constraints.
- Encode the knowledge**
 - Represent knowledge using a formal language (e.g., propositional or first-order logic).
 - Define the knowledge base (axioms, facts, rules).
- Ask queries and test**
 - Pose queries to the knowledge base.
 - Check if the inferred answers are correct.
- Debug and refine**
 - Fix incorrect or missing knowledge.
 - Add new knowledge and improve the encoding.
 - Iterate until the agent performs well on test cases.

4. Strengths and Limitations of Logic-Based AI

Strengths	Limitations
- Clear and precise semantics. - Supports sound (and often complete) inference. - Can represent complex relational knowledge. - Flexible and expressive (e.g., first-order logic). - Separation of knowledge and inference allows reuse. - Good for explanation and justification of conclusions.	- Knowledge acquisition can be difficult and time-consuming. - Requires considerable domain expertise. - Inference can be computationally expensive (especially in FOL). - Knowledge bases can become large and hard to maintain. - May not handle uncertainty (requires extensions like probabilistic logic). - Not always suitable for real-time or resource-limited environments.

6. Common Exam Questions

- Explain the Knowledge engineering process with a neat diagram.
- List and explain the steps in building a knowledge base.
- Compare reactive agents and knowledge-based agents with examples.
- Discuss the strengths and limitations of logic-based AI.
- Give applications of logical agents in the real world.
- Explain how to encode domain knowledge in first-order logic.
- Describe the inference process used by a logical agent.
- What are the challenges in knowledge acquisition and representation?
- How would you debug and refine a knowledge base?
- Which type of agent (reactive or knowledge-based) is more suitable for a given application? Justify your answer.

8. Chapter 7 Key Takeaways

- Logical agents use explicit knowledge and inference to make decisions.
- Knowledge engineering is essential for building effective logical agents.
- The process: identify task $\rightarrow$ assemble knowledge $\rightarrow$ encode $\rightarrow$ query $\rightarrow$ refine.
- Logic provides a clear and precise framework for representation and reasoning.
- Logical agents are widely applicable (planning, NLP, robotics, expert systems, etc.).
- Logic-based AI is powerful but has limitations (knowledge acquisition, scalability).
- Choosing the right agent type depends on the task and environment.

9. Quick Recap Table

Topic	Key Points
Knowledge engineering	Build KB for a logical agent (identify, assemble, encode, query, refine).
Applications	Robotics, NLP, planning, expert systems, games, etc.
Strengths	Precise, expressive, supports inference, explainable.
Limitations	Hard to acquire knowledge, computationally expensive, scalability, uncertainty.
Reactive vs. Knowledge-based	Reactive: rules, no reasoning. Knowledge-based: KB + inference, more flexible.

“ Knowledge + reasoning = intelligent behavior. The better we represent what we know, the smarter our agents can be.”